

Excerpts from Formalized Music

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Chapter V: Free Stochastic Music by Computer

After this interlude, we return to the treatment of composition by machines.

The theory put forward by *Achorripsis* had to wait four years before being realized mechanically. This realization occurred thanks to M. François Génuys of IBM-France and to M. Jacques Barraud of the Régie Autonome des Transports Parisiens.

The Paradox: Music and Computers

A Stochastic Work Executed by the IBM-7090

The general public has a number of different reactions when faced by the alliance of the machine with artistic creation. They fall into three categories:

“It is impossible to obtain a *work of art*, since by definition it is a handicraft and requires moment-by-moment “creation” for each detail and for the entire structure, while a machine is an inert thing and cannot invent.”

“Yes, one may play games with a machine or use it for speculative purposes, but the result will not be “finished”: it will represent only an experiment—interesting, perhaps, but no more.”

The enthusiasts who at the outset accept without flinching the whole frantic brouhaha of science fiction. “The moon? Well, yes, it’s within our reach. Prolonged life will also be with us tomorrow—why not a creative machine?” These people are among the credulous who, in their, idiosyncratic optimism, have replaced the myths of Icarus and the fairies, which have decayed, by the scientific civilization of the twentieth century, and science partly agrees with them. In reality, science is neither all paradox nor all animism, for it progresses in limited stages that are not foreseeable at too great a distance.

There exists in all the arts what we may call rationalism in the etymological sense: the search for proportion. The *artist* has always called upon it out of *necessity*. The rules of construction have varied widely over the centuries, but there have always been rules in every epoch because of the necessity of making oneself understood. Those who believe the first statement above are the first to refuse to apply the qualification *artistic* to a product which they do not *understand* at all.

Thus the musical scale is a convention which circumscribes the area of potentiality and permits construction within those limits in its own particular symmetry. The rules of Christian hymnography, of harmony, and of counterpoint in the various ages have allowed artists to construct and to make themselves understood by those who adopted the same constraints—through traditions, through collective taste or imitation, or through sympathetic resonance. The rules of serialism, for instance, those that banned the traditional octave doublings of tonality, imposed constraints which were partly new but none the less real.

Now everything that is rule or repeated constraint is part of the mental machine. A little “imaginary machine,” Philippot would have said—a choice, a set of decisions. A musical work can be analyzed as a multitude of mental machines. A melodic theme in a symphony is a mold, a mental machine, in the

same way as its structure is. These mental machines are something very restrictive and deterministic, and sometimes very vague and indecisive. In the last few years we have seen that this idea of mechanism is really a very general one. It flows through every area of human knowledge and action, from strict logic to artistic manifestations.

Just as the wheel was once one of the greatest products of human intelligence, a mechanism which allowed one to travel farther and faster with more luggage, so is the computer, which today allows the transformation of man's ideas. Computers resolve logical problems by heuristic methods. But computers are not really responsible for the introduction of mathematics into music; rather it is mathematics that makes use of the computer in composition. Yet if people's minds are in general ready to recognize the usefulness of geometry in the plastic arts (architecture, painting, etc.), they have only one more stream to cross to be able to conceive of using more abstract, non-visual mathematics and machines as aids to musical composition, which is more abstract than the plastic arts.

To summarize:

1. The creative thought of man gives birth to mental mechanisms, which, in the last analysis, are merely sets of constraints and choices. This process takes place in all realms of thought, including the arts.
2. Some of these mechanisms can be expressed in mathematical terms.
3. Some of them are physically realizable: the wheel, motors, bombs, digital computers, analogue computers, etc.
4. Certain mental mechanisms may correspond to certain mechanisms of nature.
5. Certain mechanizable aspects of artistic creation may be simulated by certain physical mechanisms or machines which exist or may be created.
6. It happens that computers can be useful in certain ways.

Here then is the theoretical point of departure for a utilization of electronic computers in musical composition.

We may further establish that the role of the living composer seems to have evolved, on the one hand, to one of inventing schemes (previously forms) and exploring the limits of these schemes, and on the other, to effecting the scientific synthesis of the new methods of construction and of sound emission. In a short while these methods must comprise all the ancient and modern means of musical instrument making, whether acoustic or electronic, with the help, for example, of digital-to-analogue converters; these have already been used in communication studies by N. Guttman, J. R. Pierce, and M. V. Mathews of Bell Telephone Laboratories in New Jersey. Now these explorations necessitate impressive mathematical, logical, physical, and psychological impedimenta, especially computers that accelerate the mental processes necessary for clearing the way for new fields by providing immediate experimental verifications at all stages of musical construction.

Music, by its very abstract nature, is the first of the arts to have attempted the conciliation of artistic creation with scientific thought. Its industrialization is inevitable and irreversible. Have we not already seen attempts to industrialize serial and popular music by the Parisian team of P. Barbaud, P. Blanchard, and Jeanine Charbonnier, as well as by the musicological research of Hiller and Isaacson at the University of Illinois?

In the preceding chapters we demonstrated some new areas of musical creation: Poisson, Markov processes, musical games, the thesis of the minimum of constraints, etc. They are all based on mathematics and especially on the theory of probability. They therefore lend themselves to being treated and

explored by computers. The simplest and most meaningful scheme is one of minimum constraints in composition, as exemplified by *Achorripsis*.

Thanks to my friend Georges Boudouris of the C.N.R.S. I made the acquaintance of Jacques Barraud, Engineer of the Ecole des Mines, then director of the Ensemble Electroniques de Gestion de la Société des Petroles Shell-Berre, and François Génuys, agrégé in mathematics, and head of the Etudes Scientifiques Nouvelles at IBM-France. All three are scientists, yet they consented to attempt an experiment which seemed at first far-fetched—that of a marriage of music with one of the most powerful machines in the world.

In most human relations it is rarely pure logical persuasion which is important; usually the paramount consideration is material interest. Now in this case it was not logic, much less self-interest, that arranged the betrothal, but purely experiment for experiment's sake, or game for game's sake, that induced collaboration. Stochastically speaking, my venture should have encountered failure. Yet the doors were opened, and at the end of a year and a half of contacts and hard work “the most unusual event witnessed by the firm or by this musical season [in Paris]” took place on 24 May 1962 at the headquarters of IBM-France. It was a live concert presenting a work of stochastic instrumental music entitled *ST/10-1, 080262*, which had been calculated on the IBM-7090. It was brilliantly performed by the conductor C. Simonovic and his Ensemble de Musique Contemporaine de Paris. By its passage through the machine, this work made tangible a stochastic method of composition, that of the minimum of constraints and rules.

Position of the Problem

The first working phase was the drawing up of the flow chart, i.e., writing down clearly and in order the stages of the operations of the scheme of *Achorripsis*,¹ and adapting it to the machine structure. In the first chapter we set out the entire synthetic method of this minimal structure. Since the machine is an iterative apparatus and performs these iterations with extraordinary speed, the thesis had to be broken down into a sequential series of operations reiterated in loops. An excerpt from the first flow chart is shown in Fig. V-1.

The statement of the thesis of *Achorripsis* receives its first machine-oriented interpretation in the following manner:

1. *The work consists of a succession of sequences or movements each a_i seconds long.* Their durations are totally independent (asymmetric) but have a fixed mean duration, which is introduced in the form of a parameter. These durations and their stochastic succession are given by the formula

$$P_{a_i} = ce^{-ca_i} da_i \quad (\text{See Appendix I.})$$

2. *Definition of the mean density of the sounds during a_i .* During a sequence sounds are emitted from several sonic sources. If the total number of these sounds or points during a sequence is N_{a_i} , the mean density of this point-cluster is N_{a_i}/a_i sounds/sec. In general, for a given instrumental ensemble this density has limits that depend on the number of instrumentalists, the nature of their instruments, and the technical difficulties of performance. For a large orchestra the upper limit is of the order of 150 sounds/sec. The lower limit (V3) is arbitrary and positive. We choose $(V3) = 0.11$ sounds/sec. Previous experiments led us to adopt a logarithmic progression for the density sensation with a number between 2 and 3 as its base. We adopted $e = 2.71827$. Thus the densities are included between $(V3)e^0$ and $(V3)e^R$ sounds/sec., which we can draw on a

¹See *Gravesaner Blätter*, nos. 11/12 (Mainz: Ars Viva Verlag, 1957).

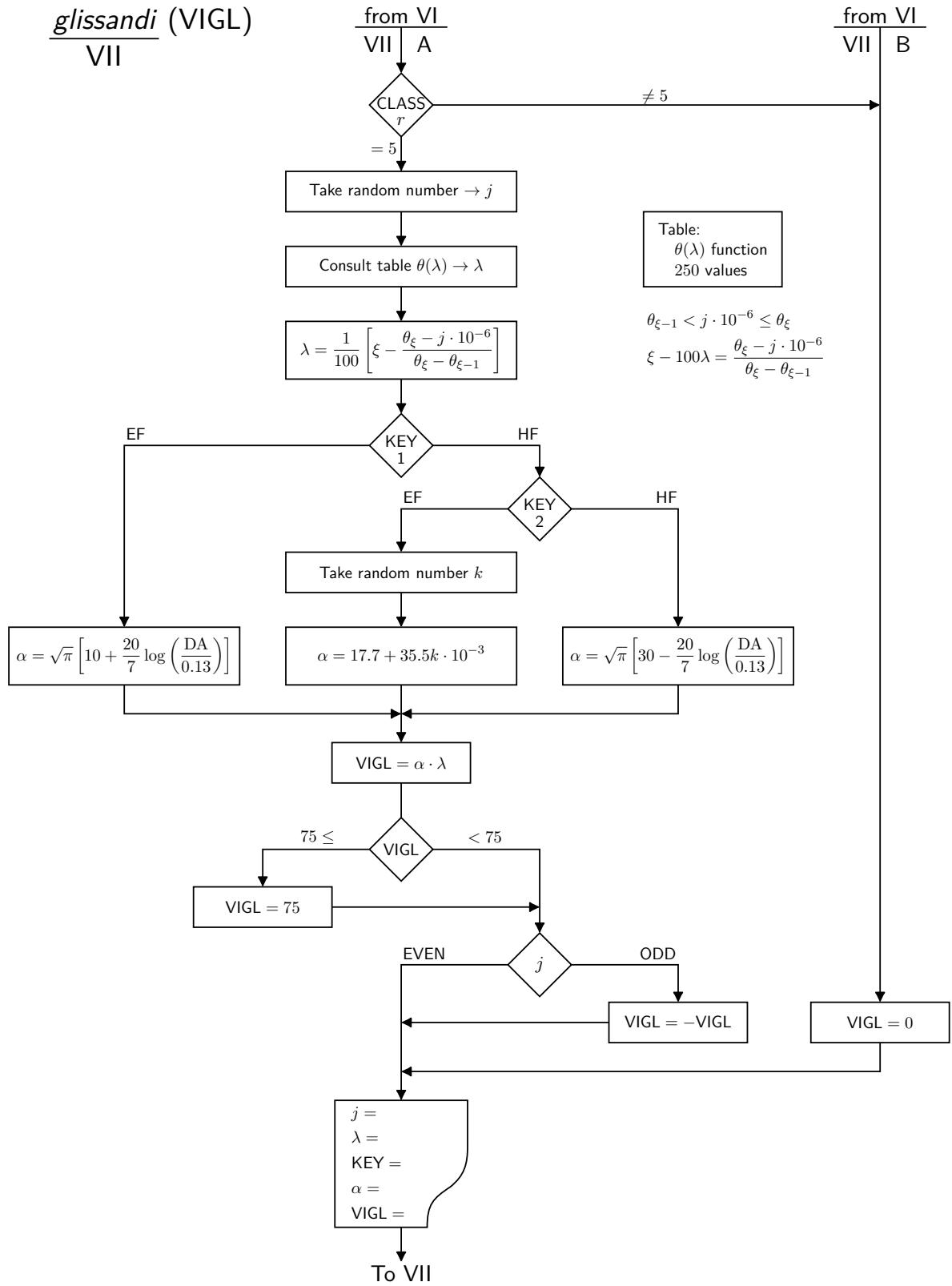


Figure 1: Excerpt from the First Flow Chart of *Achorripsis*

line graduated logarithmically (base e).² As our purpose is total independence, we attribute to each of the sequences a_i calculated in 1. a density represented by a point drawn at random from the portion of the line mentioned above. However a certain concern for continuity leads us to temper the independence of the densities among sequences a_i ; to this end we introduce a certain “memory” from sequence to sequence in the following manner:

Let a_{i-1} be a sequence of duration a_{i-1} , $(DA)_{i-1}$ its density, and a_i the next sequence with duration a_i and density $(DA)_i$. Density $(DA)_i$ will be given by the formula:

$$(DA)_i = (DA)_{i-1} e^{\pm x},$$

in which x is a segment of line drawn at random from a line segment s of length equal to $(R-0)$. The probability of x is given by

$$P_x = \frac{2}{s} \left(1 - \frac{x}{s}\right) dx \quad (\text{See Appendix I.})$$

and finally,

$$N_{a_i} = (DA)_i a_i.$$

3. *Composition Q of the orchestra during sequence a_i .* First the instruments are divided into r classes of timbres, e.g., flutes and clarinets, oboes and bassoons, brasses, bowed strings, pizzicati, col legno strokes, glissandi, wood, skin, and metal percussion instruments, etc. (See the table for *Atrées*.)

The composition of the orchestra is stochastically conceived, i.e., the distribution of the classes is not deterministic. Thus during a sequence of duration a_i it may happen that we have 80% pizzicati, 10% percussion, 7% keyboard, and 3% flute class. Under actual conditions the determining factor which would condition the composition of the orchestra is density. We therefore connect the orchestral composition with density by means of a special diagram. An example from *ST/10-1, 080262* is shown in Fig. V-2.

Fig. V-2 is expressed by the formula

$$Q_r = (n - x)(e_{n,r} - e_{n+1,r}) + e_{n,r}$$

in which r = the number of the class, $x = \log_e [(DA)_i / (V3)]$, $n = 0, 1, 2, \dots, R$, such that $n \leq x \leq n + 1$, and $e_{n,r}$ and $e_{n+1,r}$ are the probabilities of class r as a function of n . It goes without saying that the composition of this table is a precise task of great complexity and delicacy. Once these preliminaries have been completed, we can define, one after the other, the N_{a_i} sounds of sequence a_i .

4. *Definition of the moment of occurrence of the sound N within the sequence a_i .* The mean density of the points or sounds to be distributed within a_i is $k = N_{a_i} / a_i$. The formula which gives the intervals separating the sound attacks is

$$P_t = k e^{-kt} dt. \quad (\text{See Appendix I.})$$

5. *Attribution to the above sound of an instrument belonging to orchestra Q, which has already been calculated.* First class r is drawn at random with probability q_r from the orchestra ensemble calculated in 3. (Consider an urn with balls of r colors in various proportions.) Then from within class r the number of the instrument is drawn according to the probability p_n given by an arbitrary table (urn with balls of n colors). Here also the distribution of instruments within a class is delicate and complex.

² $(V3)e^R$ must be equal to the upper limit, e.g., to 150 sounds/sec. in the case of a large orchestra.

Table 1: Composition of the Orchestra for *Atrées* (ST/10-3,060962)
Timbre classes and instruments as on present input data

<i>Class</i>	<i>Timbre</i>	<i>Instrument</i>	<i>Instrument No.</i>
1	Percussion	Temple-blocks	1 — 5
		Tom-toms	6 — 9
		Maracas	10
		Susp. Cymbal	11
		Gong	12
2	Horn	French horn	1
3	Flute	Flute	1
4	Clarinet	Clarinet B $\flat$	1
		Bass clar. B $\flat$	2
5	Glissando	Violin	1
		Cello	2
		Trombone	3
6	Tremolo or flutter-tongue	Flute	1
		Clarinet B $\flat$	2
		Bass clar. B $\flat$	3
		French horn	4
		Trumpet	5
		Trombone <i>a</i>	6
		Trombone <i>b</i>	7
		(pedal notes)	
		Violin	8
		Cello	9
7	Plucked strings	Violin	1
		Cello	2
8	Struck strings*	Violin	1
		Cello	2
9	Vibraphone	Vibraphone	1
10	Trumpet	Trumpet	1
11	Trombone	Trombone <i>a</i>	1
		Trombone <i>b</i>	2
12	Bowed strings	(pedal notes)	
		Violin	1
		Cello	2

* col legno

6. *Attribution of a pitch as a function of the instrument.* Taking as the zero point the lowest B $\flat$ of the piano, we establish a chromatic scale in semitones of about 85 degrees. The range s of each instrument is thus expressed by a natural number (distance). But the pitch h_u of a sound is expressed by a decimal number of which the whole number part is related to a note of the chromatic scale within the instrument's range.

Just as for the density in 2., we accept a certain memory of or dependence on the preceding pitch

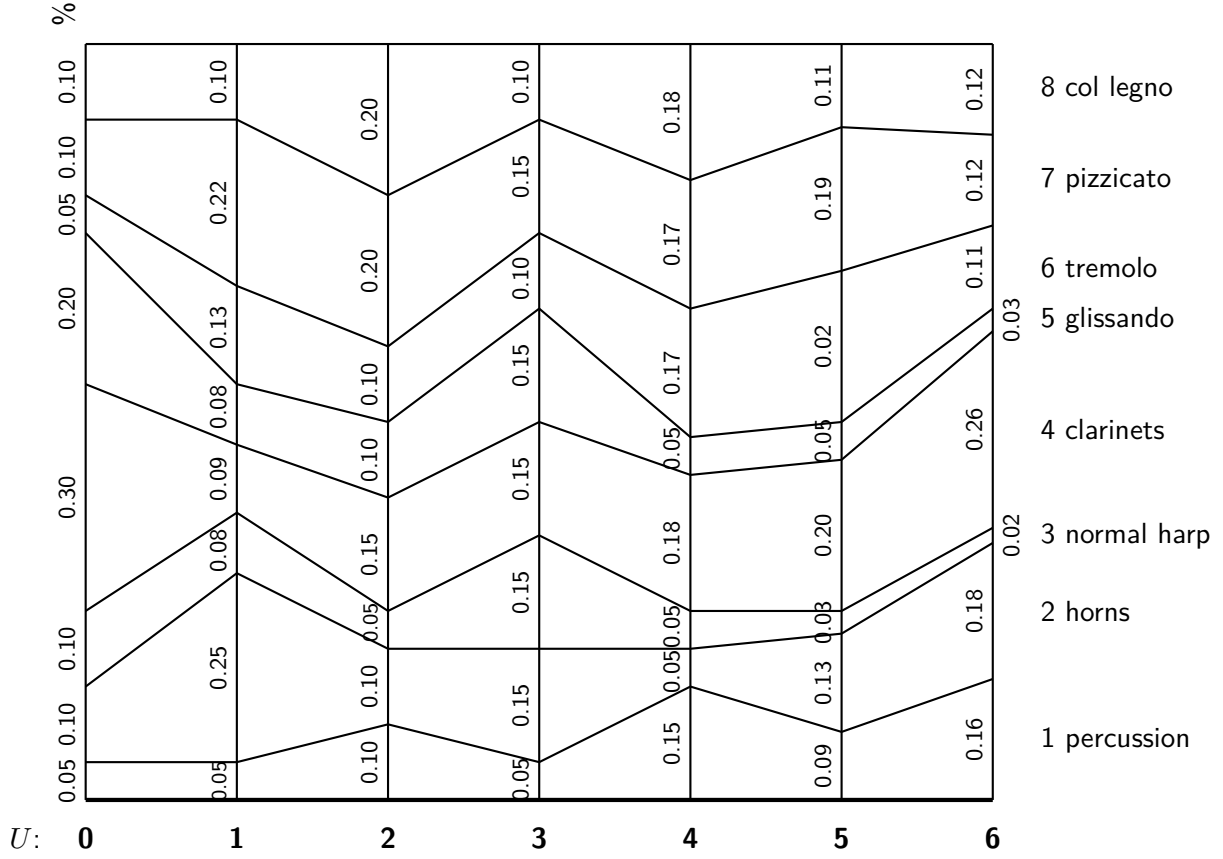


Figure 2: *ST/10-1, 080262*, Composition of the Orchestra
Density = $(DA)_i = 0.11e^U$, $U = \log_e DA/0.11$

played by the same instrument, so that we have

$$h_u = h_{u-1} \pm z,$$

where z is given by the probability formula

$$P_z = \frac{2}{s} \left(1 - \frac{z}{s}\right) dz. \quad (\text{See Appendix I.})$$

P_z is the probability of the interval z taken at random from the range s , and s is expressed as the difference between the highest and lowest pitches that can be played on the instrument.

7. *Attribution of a glissando speed if class r is characterized as a glissando.* The homogeneity hypotheses in Chap. I led us to the formula

$$f(v) = \frac{2}{a\sqrt{\pi}} e^{-v^2/a^2},$$

and by the transformation $v/a = u$ to its homologue:

$$T(u) = \frac{2}{\sqrt{\pi}} \int_0^u e^{-u^2} du,$$

for which there are tables. $f(v)$ is the probability of occurrence of the speed v (which is expressed in semitones/sec.); it has a parameter a , which is proportional to the standard deviation s ($a = s\sqrt{2}$). a is defined as a function of the logarithm of the density of sequence a_i by: an inversely proportional function

$$a = \sqrt{\pi} \left(30 - \frac{20}{R} \log [(DA)_i / (V3)] \right),$$

or a directly proportional function

$$a = \sqrt{\pi} \left(10 + \frac{20}{R} \log [(DA)_i / (V3)] \right),$$

or a function independent of density

$$a = 17.7 + 35k,$$

where k is a random number between 0 and 1.

The constants of the preceding formulae derive from the limits of the speeds that string glissandi may take.

Thus for $(DA)_i = 145$ sounds/sec.

$$a = 53.2 \text{ semitones/sec.}$$

$$2s = 75 \text{ semitones/sec.}$$

and for $(DA)_i = 0.13$ sounds/sec.

$$a = 17.7 \text{ semitones/sec.}$$

$$2s = 25 \text{ semitones/sec.}$$

8. *Attribution of a duration x to the sounds emitted.* To simplify we establish a mean duration for each instrument, which is independent of tessitura and nuance. Consequently we reserve the right to modify it when transcribing into traditional notation. The following is the list of constraints that we take into account for the establishment of duration x :

G , the maximum length of respiration or desired duration

$(DA)_i$, the density of the sequence

q_r , the probability of class r

p_n , the probability of the instrument n

Then if we define z as a parameter of a sound's duration, z could be inversely proportional to the probability of the occurrence of the instrument, so that

$$z = \frac{1}{(DA)_i p_n q_r}.$$

z will be at its maximum when $(DA)_i p_n q_r$ is at its minimum, and in this case we could choose $z_{\max} = G$.

Instead of letting $z_{\max} = G$, we shall establish a logarithmic law so as to freeze the growth of z . This law applies for any given value of z .

$$z' = G \ln z / \ln z_{\max}$$

Since we admit a total independence, the distribution of the durations x will be Gaussian:

$$f(x) = \frac{1}{s\sqrt{2\pi}} e^{-(x-m)^2/(2s^2)},$$

where m is the arithmetic mean of the durations, s the standard deviation, and

$$m - 4.25s = 0$$

$$m + 4.25s = z'$$

the linear system which furnishes us with the constants m and s . By assuming $u = (x - m)/s\sqrt{2}$ we find the function $T(u)$, for which we consult the tables.

Finally, the duration x of the sound will be given by the relation

$$x = \pm us\sqrt{2} + m.$$

We do not take into account incompatibilities between instruments, for this would needlessly burden the machine's program and calculation.

9. *Attribution of dynamic forms to the sounds emitted.* We define four zones of mean intensities: **ppp**, **p**, **f**, **ff**. Taken three at a time they yield $4^3 = 64$ permutations, of which 44 are different (an urn with 44 colors); for example, **ppp** $\leftarrow$ **f** $\rightarrow$ **p**.
10. *The same operations are begun again for each sound of the cluster N_{a_i} .*
11. *Recalculations of the same sort are made for the other sequences.*

An extract from the sequential statement was reproduced in Fig. V-1. Now we must proceed to the transcription into Fortran IV, a language "understood" by the machine (see Fig. V-3).

It is not our purpose to describe the transformation of the flow chart into Fortran. However, it would be interesting to show an example of the adaptation of a mathematical expression to machine methods.

Let us consider the elementary law of probability (density function)

$$f(x) dx = ce^{-cx} dx. \quad [20]$$

How shall we proceed in order for the computer to give us lengths x with the probability $f(x) dx$? The machine can only draw random numbers y_0 with equiprobability between 0 and 1. We shall "modulate" this probability: Assume some length x_0 ; then we have

$$\text{prob}(0 \leq x \leq x_0) = \int_0^{x_0} f(x) dx = 1 - e^{-cx_0} = F(x_0),$$

where $F(x_0)$ is the distribution function of x . But

$$F(x_0) = \text{prob}(0 \leq y \leq y_0) = y_0$$

then

$$1 - e^{-cx_0} = y_0$$

and

$$x_0 = -\frac{\ln(1 - y_0)}{c}$$

Table 2: Table of the 44 Intensity Forms Derived
from 4 Mean Intensity Values *ppp*, *p*, *f*, *ff*

<i>ppp</i> ————— <i>ppp</i>	<i>f</i> ————— <i>p</i>
<i>ppp</i> ————— <i>p</i>	<i>p</i> ————— <i>ff</i>
<i>ppp</i> < <i>p</i> > <i>ppp</i>	<i>p</i> < <i>ff</i> > <i>p</i>
<i>p</i> ————— <i>ppp</i>	<i>ff</i> ————— <i>p</i>
<i>ppp</i> ————— <i>f</i>	<i>ppp</i> < <i>ff</i> > <i>f</i>
<i>ppp</i> < <i>f</i> > <i>ppp</i>	<i>ppp</i> ————— <i>ppp</i>
<i>f</i> ————— <i>ppp</i>	<i>ppp</i> ————— <i>ppp</i>
<i>ppp</i> ————— <i>ff</i>	<i>ppp</i> ————— <i>ppp</i>
<i>ppp</i> < <i>ff</i> > <i>ppp</i>	<i>ppp</i> ————— <i>ppp</i>
<i>ff</i> ————— <i>ppp</i>	<i>ppp</i> ————— <i>ppp</i>
<i>ppp</i> < <i>f</i> > <i>p</i>	<i>ppp</i> ————— <i>ppp</i>
<i>f</i> > <i>ppp</i> < <i>p</i>	<i>ppp</i> ————— <i>ppp</i>
<i>p</i> < <i>f</i> > <i>ppp</i>	<i>ppp</i> ————— <i>ppp</i>
<i>p</i> > <i>ppp</i> < <i>f</i>	<i>ppp</i> ————— <i>ppp</i>
<i>ppp</i> < <i>ff</i> > <i>p</i>	<i>ppp</i> ————— <i>ppp</i>
<i>ff</i> > <i>ppp</i> < <i>p</i>	<i>ppp</i> ————— <i>ppp</i>
<i>p</i> > <i>ppp</i> < <i>ff</i>	<i>ppp</i> ————— <i>ppp</i>
<i>p</i> < <i>ff</i> > <i>ppp</i>	<i>ppp</i> ————— <i>ppp</i>
<i>p</i> ————— <i>p</i>	<i>ppp</i> ————— <i>ppp</i>
<i>p</i> > <i>ppp</i> < <i>p</i>	<i>ppp</i> ————— <i>ppp</i>
<i>p</i> ————— <i>f</i>	<i>ppp</i> ————— <i>ppp</i>
<i>p</i> < <i>f</i> > <i>p</i>	<i>ppp</i> ————— <i>ppp</i>

for all $x_0 \geq 0$.

Once the program is transcribed into language that the machine's internal organization can assimilate, a process that can take several months, we can proceed to punching the cards and setting up certain tests. Short sections are run on the machine to detect errors of logic and orthography and to determine the values of the entry parameters, which are introduced in the form of variables. This is a very important phase, for it permits us to explore all parts of the program and determine the modalities of its operation. The final phase is the decoding of the results into traditional notation, unless an automatic transcriber is available.

Conclusions

A large number of compositions of the same kind as *ST/10-1, 080262* is possible for a large number of orchestral combinations. Other works have already been written: *ST/48-1, 240162*, for large orchestra, commissioned by RTF (France III); *Atrées* for ten soloists; and *Morisma-Amorisima*, for four soloists.

Although this program gives a satisfactory solution to the minimal structure, it is, however, necessary

to jump to the stage of pure composition by coupling a digital-to-analogue converter to the computer. The numerical calculations would then be changed into sound, whose internal organization had been conceived beforehand. At this point one could bring to fruition and generalize the concepts described in the preceding chapters.

The following are several of the advantages of using electronic computers in musical composition:

1. The long laborious calculation made by hand is reduced to nothing. The speed of a machine such as the IBM-7090 is tremendous—of the order of 500,000 elementary operations/sec.
2. Freed from tedious calculations the composer is able to devote himself to the general problems that the new musical form poses and to explore the nooks and crannies of this form while modifying the values of the input data. For example, he may test all instrumental combinations from soloists to chamber orchestras, to large orchestras. With the aid of electronic computers the composer becomes a sort of pilot: he presses the buttons, introduces coordinates, and supervises the controls of a cosmic vessel sailing in the space of sound, across sonic constellations and galaxies that he could formerly glimpse only as a distant dream. Now he can explore them at his ease, seated in an armchair.
3. The program, i.e., the list of sequential operations that constitute the new musical form, is an objective manifestation of this form. The program may consequently be dispatched to any point on the earth that possesses computers of the appropriate type, and may be exploited by any composer pilot.
4. Because of certain uncertainties introduced in the program, the composer-pilot can instill his own personality in the sonic result he obtains.

Figure 3: Stochastic Music Rewritten in Fortran IV

C	PROGRAM FREE STOCHASTIC MUSIC (FORTRAN IV)	XEN	6
C		XEN	7
C	GLOSSARY OF THE PRINCIPAL ABBREVIATIONS	XEN	8
C			
C	A - DURATION OF EACH SEQUENCE IN SECONDS	XEN	9
C	A10,A20,A17,A35,A30 - NUMBERS FOR GLISSANDO CALCULATION	XEN	10
C	ALEA - PARAMETER USED TO ALTER THE RESULT OF A SECOND RUN WITH THE	XEN	11
C	SAME INPUT DATA	XEN	12
C	ALFA(3) - THREE EXPRESSIONS ENTERING INTO THE THREE SPEED VALUES	XEN	13
C	OF THE SLIDING TONES (GLISSANDI)	XEN	14
C	ALIM - MAXIMUM LIMIT OF SEQUENCE DURATION A	XEN	15
C	(AMAX(I),I=1,KTR) TABLE OF AN EXPRESSION ENTERING INTO THE	XEN	16
C	CALCULATION OF THE NOTE LENGTH IN PART 8	XEN	17
C	BF - DYNAMIC FORM NUMBER. THE LIST IS ESTABLISHED INDEPENDENTLY	XEN	18
C	OF THIS PROGRAM AND IS SUBJECT TO MODIFICATION	XEN	19
C	DELTA - THE RECIPROCAL OF THE MEAN DENSITY OF SOUND EVENTS DURING	XEN	20
C	A SEQUENCE OF DURATION A	XEN	21
C	(E(I,J),I=1,KTR,J=1,KTE) - PROBABILITIES OF THE KTR TIMBRE CLASSES	XEN	22
C	INTRODUCED AS INPUT DATA, DEPENDING ON THE CLASS NUMBER I=KR AND	XEN	23
C	ON THE POWER J=U OBTAINED FROM $V3 \cdot \exp(U) = DA$	XEN	24
C	EPSI - EPSILON FOR ACCURACY IN CALCULATING PN AND E(I,J), WHICH	XEN	25
C	IT IS ADVISABLE TO RETAIN.	XEN	26
C	(GN(I,J),I=1,KTR,J=1,KTS) - TABLE OF THE GIVEN LENGTH OF BREATH	XEN	27
C	FOR EACH INSTRUMENT, DEPENDING ON CLASS I AND INSTRUMENT J	XEN	28
C	GTNA - GREATEST NUMBER OF NOTES IN THE SEQUENCE OF DURATION A	XEN	29
C	GTNS - GREATEST NUMBER OF NOTES IN KW LOOPS	XEN	30
C	(HAMIN(I,J),HAMAX(I,J),HBMIN(I,J),HBMAX(I,J),I=1,KTR,J=1,KTS)	XEN	31
C	TABLE OF INSTRUMENT COMPASS LIMITS, DEPENDING ON TIMBRE CLASS I	XEN	32
C	AND INSTRUMENT J. TEST INSTRUCTION 480 IN PART 6 DETERMINES	XEN	33
C	WHETHER THE HA OR THE HB TABLE IS FOLLOWED. THE NUMBER 7 IS	XEN	34
C	ARBITRARY.	XEN	35
C	JW - ORDINAL NUMBER OF THE SEQUENCE COMPUTED.	XEN	36
C	KNL - NUMBER OF LINES PER PAGE OF THE PRINTED RESULT.KNL=50	XEN	37
C	KR1 - NUMBER IN THE CLASS KR=1 USED FOR PERCUSSION OR INSTRUMENTS	XEN	38
C	WITHOUT A DEFINITE PITCH.	XEN	39
C	KTE - POWER OF THE EXPONENTIAL COEFFICIENT E SUCH THAT	XEN	40
C	$DA(MAX) = V3 \cdot (E^{**}(KTE-1))$	XEN	41
C	KTR - NUMBER OF TIMBRE CLASSES	XEN	42
C	KW - MAXIMUM NUMBER OF JW	XEN	43
C	KTEST1,TAV1,ETC - EXPRESSIONS USEFUL IN CALCULATING HOW LONG THE	XEN	44
C	VARIOUS PARTS OF THE PROGRAM WILL RUN.	XEN	45
C	KT1 - ZERO IF THE PROGRAM IS BEING RUN, NONZERO DURING DEBUGGING	XEN	46
C	KT2 - NUMBER OF LOOPS, EQUAL TO 15 BY ARBITRARY DEFINITION.	XEN	47
C	(MODI(IX8),IX8=7,1) AUXILIARY FUNCTION TO INTERPOLATE VALUES IN	XEN	48
C	THE TETA(256) TABLE (SEE PART 7)	XEN	49
C	NA - NUMBER OF SOUNDS CALCULATED FOR THE SEQUENCE A(NA=DA*A)	XEN	50
C	(NT(I),I=1,KTR) NUMBER OF INSTRUMENTS ALLOCATED TO EACH OF THE	XEN	51
C	KTR TIMBRE CLASSES.	XEN	52
C	(PN(I,J),I=1,KTR,J=1,KTS),(KTS=NT(I),I=1,KTR) TABLE OF PROBABILITY	XEN	53
C	OF EACH INSTRUMENT OF THE CLASS I.	XEN	54
C	(Q(I),I=1,KTR) PROBABILITIES OF THE KTR TIMBRE CLASSES, CONSIDERED	XEN	55
C	AS LINEAR FUNCTIONS OF THE DENSITY DA.	XEN	56
C	(S(I),I=1,KTR) SUM OF THE SUCCESSIVE Q(I) PROBABILITIES, USED TO	XEN	57
C	CHOOSE THE CLASS KR BY COMPARING IT TO A RANDOM NUMBER X1 (SEE	XEN	58
C	PART 3, LOOP 380 AND PART 5, LOOP 430).	XEN	59

C	SINA - SUM OF THE COMPUTED NOTES IN THE NEW CLOUDS NA, ALWAYS LESSXEN	60
C	THAN GTNS (SEE TEST IN PART 10).	XEN 61
C	SQPI - SQUARE ROOT OF PI (3.14159...)	XEN 62
C	TA - SOUND ATTACK TIME ABCISSA.	XEN 63
C	TETA(256) - TABLE OF THE 256 VALUES OF THE INTEGRAL OF THE NORMAL XEN	64
C	DISTRIBUTION CURVE WHICH IS USEFUL IN CALCULATING GLISSANDO SPEED XEN	65
C	AND SOUND EVENT DURATION.	XEN 66
C	VIGL - GLISSANDO SPEED (VITESSE GLISSANDO), WHICH CAN VARY AS, BE XEN	67
C	INDEPENDENT OF, OR VARY INVERSELY AS THE DENSITY OF THE SEQUENCE, XEN	68
C	THE ACTUAL MODE OF VARIATION EMPLOYED REMAINING THE SAME FOR THE XEN	69
C	ENTIRE SEQUENCE (SEE PART 7).	XEN 70
C	VITLIM - MAXIMUM LIMITING GLISSANDO SPEED (IN SEMITONES/SEC), XEN	71
C	SUBJECT TO MODIFICATION.	XEN 72
C	V3 - MINIMUM CLOUD DENSITY DA	XEN 73
C	(Z1(I),Z2(I),I=1,8) TABLE COMPLEMENTARY TO THE TETA TABLE.	XEN 74
C		XEN 75

C			XEN	76
C	READ	CONSTANTS AND TABLES	XEN	78
C			XEN	77
	DIMENSION	Q(12),S(12),E(12,12),PN(12,50),SPN(12,50),NT(12),	XEN	79
	*HAMIN(12,50),HAMAX(12,50),HBMIN(12,50),HBMAX(12,50),GN(12,50),H(12	XEN	80	
	*,50),TETA(256),VIGL(3),MODI(7),Z1(8),Z2(8),ALFA(3),AMAX(12)	XEN	81	
C			XEN	82
C			XEN	83
C			XEN	84
	I=1		XEN	85
	DO 10 IX=1,7		XEN	86
	IX8=8-IX		XEN	87
	MODI(IX8)=I		XEN	88
10	I=I+1		XEN	89
C			XEN	90
	READ 20,(TETA(I),I=1,256)		XEN	91
20	FORMAT(12F6.6)		XEN	92
	READ 30,(Z1(I),Z2(I),I=1,8)		XEN	93
30	FORMAT(6(F3.2,F9.8)/F3.2,F9.8,E6.2,F9.8)		XEN	94
	PRINT 40,TETA,Z1,Z2		XEN	95
40	FORMAT('1THE TETA TABLE=',/,21(12F10.6,/),4F10.6,/////,		XEN	96
	* 'THE Z1 TABLE=',/,7F6.2,E12.3,///, 'THE Z2 TABLE=',/,8F14.8,/	XEN	97	
	*,1H1)		XEN	98
	READ 50,DELTA,V3,A10,A20,A17,A30,A35,BF,SQPI,EPSI,VITLIM,ALEA,	AXEN	99	
	*LIM		XEN	100
50	FORMAT(F3.0,F3.3,5F3.1,F2.0,F8.7,F8.8,F4.2,F8.8,F5.2)		XEN	114
	READ 60,KT1,KT2,KW,KNL,KTR,KTE,KR1,GTNA,GTNS,(NT(I),I=1,KTR)		XEN	115

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60  FORMAT(5I3,2I2,2F6.0,12I2)                                XEN 126
    PRINT 70,DELTA,V3,A10,A20,A17,A30,A35,BF,SQPI,EPSI,VITLIM,ALEA, AXEN 127
    *LIM,KT1,KT2,KW,KNL,KTR,KTE,KR1,GTNA,GTNS,((I,NT(I)),I=1,KTR) XEN 128
70  FORMAT('1DELTA= ',F4.0/, 'V3= ',F6.3/, 'A10= ',F4.1/,      XEN 129
    * 'A20= ',F4.1/, 'A17= ',F4.1/, 'A30= ',F4.1/, 'A35= ',F4.1, XEN 130
    *, 'BF= ',F3.0/, 'SQPI= ',F11.8/, 'EPSI= ',F12.8/, 'VITLIM= ' XEN 131
    *,F5.2/, 'ALEA= ',F12.8/, 'ALIM= ',F6.2/, 'KT1= ',I3/,      XEN 132
    * 'KT2= ',I3/, 'KW= ',I3/, 'KNL= ',I3/, 'KTR= ',I3/,        XEN 133
    * 'KTE= ',I2/, 'KR1= ',I2/, 'GTNA= ',F7.0/, 'GTNS= ',F7.0,  XEN 134
    */ ,12('INCLASS',I2, 'THEREARE',I2, 'INSTRUMENTS.',/))      XEN 135
    READ 80,KTEST3,KTEST1,KTEST2                                XEN 136
80  FORMAT(5I3)                                                XEN 141
    PRINT 90,KTEST3,KTEST1,KTEST2                                XEN 142
90  FORMAT('KTEST3= ',I3/, 'KTEST1= ',I3/, 'KTEST2= ',I3)      XEN 143
C                                                                    XEN 144
    IF(KTEST3.NE.0) PRINT 830                                    XEN 145
    R=KTE-1                                                      XEN 146
    A10=A10*SQPI                                                  XEN 147
    A20=A20*SQPI/R                                              XEN 148
    A30=A30*SQPI                                                  XEN 149
C    IF ALEA IS NON-ZERO,THE RANDOM NUMBER IS GENERATED FROM THE TIME XEN 150
C    WHEN THE FOLLOWING INSTRUCTION IS EXECUTED. IF ALEA IS NON-ZERO XEN 151
C    EACH RUN OF THIS PROGRAM WILL PRODUCE DIFFERENT OUTPUT DATA. XEN 152
    IF(ALEA.NE.0.0) CALL RANFSET(TIMEF(1))                       XEN 153
    PRINT 830                                                    XEN 154
    DO 130 I=1,KTR                                              XEN 155
    Y=0.0                                                         XEN 156
    KTS=NT(I)                                                    XEN 157
    READ 100,(HAMIN(I,J),HAMAX(I,J),HBMIN(I,J),HBMAX(I,J),GN(I,J), XEN 158
    *PN(I,J),J=1,KTS)                                           XEN 159
100  FORMAT(5(5F2.0,F3.3))                                       XEN 160
    PRINT 110,I,(J,HAMIN(I,J),HAMAX(I,J),HBMIN(I,J),HBMAX(I,J),GN(I,J) XEN 161
    *,PN(I,J),J=1,KTS)                                           XEN 162
110  FORMAT(////, 'INCLASSNUMBER',I2/,('FORINSTRUMENTNO.',I2, XEN 163
    * 'HAMIN= ',F3.0, 'HAMAX= ',F3.0, 'HBMIN= ',F3.0, 'HBMAX= ', XEN 164
    * F3.0, 'GN= ',F3.0, 'ANDPN= ',F6.3))                       XEN 165
    DO 120 J=1,KTS                                              XEN 166
    Y=Y+PN(I,J)                                                  XEN 167
120  SPN(I,J)=Y                                                  XEN 168
130  IF(ABSF(Y-1.0).GE.EPSI) CALL EXIT                          XEN 169
C                                                                    XEN 170
    DO 150 I=1,KTR                                              XEN 171
    READ 140,(E(I,J),J=1,KTE)                                    XEN 172
140  FORMAT(12F2.2)                                             XEN 173
150  PRINT 160,I,(J,E(I,J),J=1,KTE)                             XEN 174
160  FORMAT(////////, 'CLASSNUMBER',I2/,('INDENSITYLEVEL',I2, XEN 175
    * 'HASAPPROBABILITYOF',F6.2))                               XEN 176
    DO 180 J=1,KTE                                              XEN 177
    Y=0.0                                                         XEN 178
    DO 170 I=1,KTR                                              XEN 179
170  Y=Y+E(I,J)                                                  XEN 180
180  IF(ABSF(Y-1.0).GE.EPSI) CALL EXIT                          XEN 181
    DO 200 I=1,KTR                                              XEN 182
    AMAX(I)=1.0/E(I,1)                                           XEN 183
    DO 200 J=2,KTE                                              XEN 184
    AJ=J-1                                                        XEN 185
    AX=1.0/(E(I,J)*EXPJ(AJ))                                     XEN 186
    IF (KT1.NE.0) PRINT 190,AX                                    XEN 187
190  FORMAT(1H ,9E12.8)                                         XEN 188

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[illegible]

! C	XEN	241
! 360 SINA=SINA + FLOATF(NA)	XEN	242
! XLOGDA=U	XEN	243
! XALOG=A20 *XLOGDA	XEN	244
! M=XINTF(XLOGDA)	XEN	245
! IF ((M+2).GT.KTE) M=KTE-2	XEN	246
! SR=0.0	XEN	247
! M1=M+1	XEN	248
! M2=M+2	XEN	249
! DO 380 I=1,KTR	XEN	250
! ALFX=E(I,M1)	XEN	251
! BETA=E(I,M2)	XEN	252
! XM=M	XEN	253
! QR=(XLOGDA-XM) * (BETA-ALFX) + ALFX	XEN	254
! IF (KT1.NE.0) PRINT 370,XM,ALFX,BETA	XEN	255
! 370 FORMAT(1H ,3F20.8)	XEN	256
! Q(I)=QR	XEN	257
! SR=SR+QR	XEN	258
! 380 S(I)=SR	XEN	259
! IF (KT1.NE.0) PRINT 390,(Q(I),I=1,KTR),(S(I),I=1,KTR)	XEN	260
! 390 FORMAT(1H ,12F9.4)	XEN	261
! C	XEN	262
! C PART 4,DEFINE INSTANT TA OF EACH POINT IN SEQUENCE A	XEN	263
! C	XEN	264
! IF (KTEST2.NE.0) TAV2=TIMEF(1)	XEN	265
! N=1	XEN	266
! T=0.0	XEN	267
! TA=0.0	XEN	268
! GO TO 410	XEN	269
! 400 N=N+1	XEN	270
! X=RANF(-1)	XEN	271
! T=-LOGF(X)/DA	XEN	272
! TA=TA+T	XEN	273
! 410 IF (KT1.NE.0) PRINT 420,N,X,T,TA	XEN	274
! 420 FORMAT(//,I8,3E20.8)	XEN	275
! C	XEN	276
! C PART 5,DEFINE CLASS AND INSTRUMENT NUMBER TO EACH POINT OF A	XEN	277
! C	XEN	278
! X1=RANF(-1)	XEN	279
! DO 430 I=1,KTR	XEN	280
! 430 IF (X1.LE.S(I)) GO TO 440	XEN	281
! I=KTR	XEN	282
! 440 KTS=NT(I)	XEN	283
! KR=I	XEN	284
! X2=RANF(-1)	XEN	285
! DO 450 J=1,KTS	XEN	286
! SPIEN=SPN(KR,J)	XEN	287
! INSTRM=J	XEN	288
! 450 IF (X2.LE.SPIEN) GO TO 460	XEN	289
! INSTRM=KTS	XEN	290
! 460 PIEN=PN(KR,INSTRM)	XEN	291
! IF (KT1.NE.0) PRINT 470,X1,S(KR),KR,X2,SPIEN,INSTRM	XEN	292
! 470 FORMAT(1H ,2E20.8,I6,2E20.8,I6)	XEN	293
! C	XEN	294
! C PART 6,DEFINE PITCH HN FOR EACH POINT OF SEQUENCE A	XEN	295
! C	XEN	296
! IF (KR.GT.1) GO TO 480	XEN	297
! IF (INSTRM.GE.KR1) GO TO 490	XEN	298
! HX=0.0	XEN	299

!	GO TO 560	XEN 300
!	480 IF (KR.LT.7) GO TO 490	XEN 301
!	HSUP=HBMAX(KR, INSTRM)	XEN 302
!	HINF=HBMIN(KR, INSTRM)	XEN 303
!	GO TO 500	XEN 304
!	490 HSUP=HAMAX(KR, INSTRM)	XEN 305
!	HINF=HAMIN(KR, INSTRM)	XEN 306
!	500 HM=HSUP-HINF	XEN 307
!	HPR=H(KR, INSTRM)	XEN 308
!	K=0	XEN 309
!	IF (HPR.LE.0.0) GO TO 520	XEN 310
!	510 X=RANF(-1)	XEN 311
!	IF (N.GT.1) GO TO 530	XEN 312
!	520 HX=HINF+HM*X RANF(-1)	XEN 313
!	GO TO 560	XEN 314
!	530 IF(RANF(-1).GE.0.5) GO TO 540	XEN 315
!	HX=HPR+HM * (1.0-SQRTF(X))	XEN 316
!	GO TO 550	XEN 317
!	540 HX=HPR-HM * (1.0-SQRTF(X))	XEN 318
!	550 IF((HX.GE.HINF).AND.(HX.LE.HSUP)) GO TO 560	XEN 319
!	IF (K.GE.KT2) GO TO 520	XEN 320
!	K=K+1	XEN 321
!	GO TO 510	XEN 322
!	560 H(KR, INSTRM)=HX	XEN 323
!	IF (KT1.NE.0) PRINT 570,K,X,HX	XEN 324
!	570 FORMAT(1H ,I6,2E20.8)	XEN 325
!	C	XEN 326
!	C PART 7,DEFINE SPEED VIGL TO EACH POINT OF A	XEN 327
!	C	XEN 328
!	IF (KR.EQ.5) GO TO 580	XEN 329
!	VIGL(1)=0.0	XEN 330
!	VIGL(2)=0.0	XEN 331
!	VIGL(3)=0.0	XEN 332
!	X1=0.0	XEN 333
!	X2=0.0	XEN 334
!	XLAMBDA=0.0	XEN 335
!	GO TO 740	XEN 336
!	580 KX=1	XEN 337
!	590 X1=RANF(-1)	XEN 338
!	IF (X1-0.9997) 600,650,680	XEN 339
!	600 I=128	XEN 340
!	DO 630 IX=1,7	XEN 341
!	IF(TETA(I)-X1) 610,640,620	XEN 342
!	610 I=I+MODI(IX)	XEN 343
!	GO TO 630	XEN 344
!	620 I=I-MODI(IX)	XEN 345
!	630 CONTINUE	XEN 346
!	IF(TETA(I)-X1) 670,640,660	XEN 347
!	640 XLAMBDA=FLOATF(I-1)/100.0	XEN 348
!	GO TO (720,760), KX	XEN 349
!	650 XLAMBDA=2.55	XEN 350
!	GO TO (720,760), KX	XEN 351
!	660 I=I-1	XEN 352
!	670 TX1=TETA(I)	XEN 353
!	XLAMBDA=(FLOATF(I-1)+(X1-TX1)/(TETA(I+1)-TX1))/100.0	XEN 354
!	GO TO (720,760), KX	XEN 355
!	680 DO 690 I=2,7	XEN 356
!	TX1=Z2(I)	XEN 357
!	IF(X1-TX1) 700,710,690	XEN 358

!	690	CONTINUE	XEN	359
!		I=8	XEN	360
!		TX1=1.0	XEN	361
!	700	TX2=Z1(I)	XEN	362
!		XLAMBDA=TX2-((TX1-X1)/(TX1-Z2(I-1)))*(TX2-Z1(I-1))	XEN	363
!		GO TO (720,760), KX	XEN	364
!	710	XLAMBDA=Z1(I)	XEN	365
!		GO TO(720,760), KX	XEN	366
!	720	ALFA(1)=A10+XALOG	XEN	367
!		ALFA(3)=A30-XALOG	XEN	368
!		X2=RANF(-1)	XEN	369
!		ALFA(2)=A17+A35*X2	XEN	370
!		DO 730 I=1,3	XEN	371
!		VIGL(I)=INTF(ALFA(I)*XLAMBDA+0.5)	XEN	372
!		IF (VIGL(I).LT.0.0) VIGL(I)=-VIGL(I)	XEN	373
!		IF (VIGL(I).GT.VITLIM) VIGL(I)=VITLIM	XEN	374
!	730	IF (RANF(-1).LT.0.5) VIGL(I)=-VIGL(I)	XEN	375
!	740	IF(KT1.NE.0) PRINT 750,X1,X2,XLAMBDA,VIGL	XEN	376
!	750	FORMAT(1H ,6E19.8)	XEN	377
!	C		XEN	378
!	C	PART 8,DEFINE DURATION FOR EACH POINT OF A	XEN	379
!	C		XEN	380
!		IF ((KR.EQ.7).OR.(KR.EQ.8)) GO TO 780	XEN	381
!		ZMAX=AMAX(KR)/(V3*PIEN)	XEN	382
!		G=GN(KR,INSTRM)	XEN	383
!		RO=G/LOGF(ZMAX)	XEN	384
!		QPND=1.0/(Q(KR)*PIEN*DA)	XEN	385
!		GE=ABSF(RO*LOGF(QPND))	XEN	386
!		XMU=GE/2.0	XEN	387
!		SIGMA=GE/4.0	XEN	388
!		KX=2	XEN	389
!		GO TO 590	XEN	390
!	760	TAU=SIGMA*XLAMBDA*1.4142	XEN	391
!		X2=RANF(-1)	XEN	392
!		IF (X2.GE.0.5) GO TO 770	XEN	393
!		XDUR=XMU+TAU	XEN	394
!		GO TO 790	XEN	395
!	770	XDUR=XMU-TAU	XEN	396
!		IF (XDUR.GE.0.0) GO TO 790	XEN	397
!	780	XDUR=0.0	XEN	398
!	790	IF(KT1.NE.0)PRINT 800,ZMAX,XMU,SIGMA,X1,XLAMBDA,X2,XDUR	XEN	399
!	800	FORMAT(1H ,5E15.8,E11.4,E15.8)	XEN	400
!	C		XEN	401
!	C	PART 9,DEFINE INTENSITY FORM TO EACH POINT OF A	XEN	402
!	C		XEN	403
!		IFORM=XINTF(RANF(-1)*BF+0.5)	XEN	404
!		IF (KT1.EQ.0) GO TO 840	XEN	405
!		IF (NLINE.LT.KNL) GO TO 810	XEN	406
!		IF (NLINE.EQ.KNL) GO TO 820	XEN	407
!		NLINE=1	XEN	408
!		GO TO 900	XEN	409
!	810	NLINE=NLINE+1	XEN	410
!		GO TO 900	XEN	411
!	820	PRINT 830	XEN	412
!	830	FORMAT(1H1)	XEN	413
!		NLINE=0	XEN	414
!		GO TO 900	XEN	415
!	840	IF (NLINE.GE.KNL) GO TO 850	XEN	416
!		NLINE=NLINE+1	XEN	417

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!      GO TO 880
!      850 PRINT 860,JW,A,NA,(Q(I),I=1,KTR)
!      860 FORMAT('1□□JW=',I3,4X,'A=',F8.2,4X,'NA=',I6,4X,'Q(I)=' ,12(F4.2,'/'
!      *) ,//)
!      PRINT 870
!      870 FORMAT(6X,'N',8X,'START',5X,'CLASS',4X,'INSTRM',4X,'PITCH',6X,
!      *'GLISS1',4X,'GLISS2',4X,'GLISS3',8X,'DURATION',5X,'DYNAM')
!      NLINE=1
!      880 PRINT 890,N,TA,KR,INSTRM,HX,(VIGL(I),I=1,3),XDUR,IFORM
!      890 FORMAT(1H ,I7,F12.2,I9,I8,F11.1,F13.1,2F10.1,F14.2,I11)
! C
! C      PART 10,REPEAT SAME DEFINITIONS FOR ALL POINTS OF A
! C
!      900 IF (N.LT.NA) GO TO 400
! C
! C      PART 11, REPEAT SEQUENCES A
! C
!      IF (KTEST2.EQ.0) GO TO 910
!      TAP2=TIMEF(1)-TAV2
!      TAP2=TAP2/FLOATF(NA)
!      PRINT 750,TAP2
! C
!      910 IF (JW.GE.KW) GO TO 930
!      920 JW=JW+1
!      IF (GTNS.GT.SINA) GO TO 220
!      930 IF (KTEST1.EQ.0) CALL EXIT
!      940 TAP1=TIMEF(-1)-TAV1
!      TAP1=TAP1/FLOATF(KW)
!      PRINT 750,TAP1
! C
!      END

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XEN 418
XEN 419
XEN 420
XEN 421
XEN 422
XEN 423
XEN 424
XEN 425
XEN 426
XEN 427
XEN 428
XEN 429
XEN 430
XEN 431
XEN 432
XEN 433
XEN 434
XEN 435
XEN 436
XEN 437
XEN 438
XEN 439
XEN 440
XEN 441
XEN 442
XEN 443
XEN 444
XEN 445
XEN 446
XEN 447
XEN 448

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C
C      DATA FOR ATREES (ST/10-3, 060962)
C

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00003467005000000154800500
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0000336310999
00001953108000000101307200
00003487155000000157215500
25080408011309
08071602010110
03030420010110
02050325010112
03350315011505
02100302103907
02020203150207
0202020410207
03090317041609
03132003200509
02052801030409
45011202020106

```

Figure 4: Provisional Results of One Phase of the Analysis

JW= 1 A= 9.13 NA= 95
Q(1)=0.12/0.04/0.04/0.05/0.12/0.29/0.04/0.04/0.14/0.06/0.06/0.03/

N	START	CLASS	INSTRM	PITCH	GLISS1	GLISS2	GLISS3	DURATION	DYNAM
1	0.00	7	1	34.0	0.0	0.0	0.0	0.00	3
2	0.10	10	1	43.2	0.0	0.0	0.0	0.41	50
3	0.11	6	8	81.3	0.0	0.0	0.0	0.63	21
4	0.13	6	3	47.0	0.0	0.0	0.0	0.18	10
5	0.18	1	4	0.0	0.0	0.0	0.0	1.90	29
6	0.25	9	1	48.7	0.0	0.0	0.0	1.51	35
7	0.33	6	7	11.4	0.0	0.0	0.0	0.37	42
8	0.34	9	1	38.1	0.0	0.0	0.0	0.00	59
9	0.40	1	1	0.0	0.0	0.0	0.0	2.20	45
10	0.41	6	9	55.0	0.0	0.0	0.0	1.07	0
11	0.76	6	7	11.5	0.0	0.0	0.0	0.40	7
12	0.90	8	2	23.2	0.0	0.0	0.0	0.00	19
13	1.00	7	2	26.9	0.0	0.0	0.0	0.00	6
14	1.09	10	1	46.2	0.0	0.0	0.0	0.32	57
15	1.09	6	2	68.5	0.0	0.0	0.0	0.71	25
16	1.23	6	3	46.9	0.0	0.0	0.0	0.64	32
17	1.42	6	1	44.0	0.0	0.0	0.0	0.44	1
18	1.57	10	1	36.2	0.0	0.0	0.0	0.22	21
19	1.65	4	2	32.5	0.0	0.0	0.0	1.09	13

20	1.78	6	8	72.6	0.0	0.0	0.0	0.06	60
21	1.92	6	3	38.9	0.0	0.0	0.0	0.55	60
22	1.94	5	1	74.6	71.0	-25.0	-71.0	0.80	62
23	2.18	4	1	32.6	0.0	0.0	0.0	1.50	50
24	2.18	6	6	50.9	0.0	0.0	0.0	0.60	26
25	2.19	1	12	0.0	0.0	0.0	0.0	4.58	24
26	2.20	9	1	49.3	0.0	0.0	0.0	0.02	58
27	2.23	9	1	51.0	0.0	0.0	0.0	0.22	13
28	2.32	7	1	36.9	0.0	0.0	0.0	0.00	43
29	2.33	4	1	31.8	0.0	0.0	0.0	1.38	56
30	2.54	1	6	0.0	0.0	0.0	0.0	0.28	14
31	2.57	11	2	12.2	0.0	0.0	0.0	1.69	40
32	2.71	9	1	48.5	0.0	0.0	0.0	0.37	55
33	2.80	1	5	0.0	0.0	0.0	0.0	1.50	58
34	3.28	5	2	15.4	49.0	5.0	-31.0	0.52	21
35	3.33	1	7	0.0	0.0	0.0	0.0	1.38	8
36	3.38	5	2	47.3	-71.0	-17.0	46.0	1.05	4
37	3.55	10	1	37.6	0.0	0.0	0.0	0.14	24
38	3.56	1	9	0.0	0.0	0.0	0.0	1.30	0
39	3.60	9	1	64.3	0.0	0.0	0.0	0.19	12
40	3.64	12	2	52.2	0.0	0.0	0.0	3.72	9
41	3.65	6	5	59.0	0.0	0.0	0.0	0.83	28
42	3.71	5	3	38.8	25.0	2.0	-15.0	0.00	11
43	3.80	6	8	75.6	0.0	0.0	0.0	0.43	17
44	3.87	6	2	51.5	0.0	0.0	0.0	0.77	57
45	3.89	6	7	12.1	0.0	0.0	0.0	0.39	2
46	4.15	5	2	43.0	-71.0	24.0	71.0	1.16	2
47	4.15	5	1	80.3	36.0	4.0	22.0	0.85	50
48	4.25	9	1	59.9	0.0	0.0	0.0	0.10	10
49	4.31	12	2	40.1	0.0	0.0	0.0	2.49	33
50	4.33	1	10	0.0	0.0	0.0	0.0	0.46	34

Appendix I

Two Laws of Continuous Probability

First Law

$$P_x = ce^{-cx} dx.$$

Let OA be a segment of a straight line of length ℓ on which we place n points. Their linear density is $c = n/\ell$. Suppose that ℓ and n increase indefinitely while c remains constant. Suppose also that these points are numbered $A_1, A_p, A_q, \dots$ and are distributed from left to right beginning at the origin 0. Let

$$x_1 = A_1A_p, \quad x_2 = A_pA_q, \quad x_3 = A_qA_r, \dots, x_i = A_sA_t.$$

The probability that the i -th segment will have a length x_i between x and $x + dx$ is

$$P_x = ce^{-cx} dx.$$

Now the probability p_n , that there will be n points on a segment x is given by the recurrence formula

$$\frac{p_{n+1}}{p_n} = \frac{cx}{n+1}.$$

Therefore $p_1 = (cx/1)p_0$. But $p_0 = e^{-cx}$ and

$$e^{-cx} = 1 - \frac{cx}{1!} + \frac{(cx)^2}{2!} - \frac{(cx)^3}{3!} + \dots.$$

If x is very small and if we denote it by dx , we have

$$p_0 = 1 - cdx + \frac{c^2(dx)^2}{2!} - \dots.$$

Since the powers of dx are infinitely small for high values, $p_0 = 1 - cdx$ and $p_1 = cdxp_0 = cdx$. Hence, the probability P_x is composed of the probability $p_0 = e^{-cx}$, that there will be no point on the segment x , and the probability $p_1 = cdx$, that there will be a point in dx .

Approximate Calculation of the Same Probability (For Calculation by Hand)

Let there be d points to be placed on a straight line of length ℓ . The linear density is $c = d/\ell$ points on length ℓ . If the lengths are expressed in units v then $\ell = av$ ($a > 0$) and $cv = d/a$ points in the unit of length v .

Then $x_i = iv$ ($i = 0, 1, 2, 3, \dots$), and the probability, the asymptotic limit of the relative frequency of the segment x_i , will be

$$P_{x_i} = ce^{-civ} \Delta x_i. \quad (1)$$

We shall now define the quantity Δx_i . The probability (??) is composed of the probability $p_0 = e^{-civ}$, that there will be no point on x_i , and the probability $p_1 = c\Delta x_i$ that there will be a point in Δx_i if $(c\Delta x_i)^2$ is small enough to be ignored. Set

$$0 < (c\Delta x_i)^2 < w10^{-n},$$

where n is a sufficiently large natural number; this expression becomes

$$0 < \Delta x_i < c^{-1} \cdot 10^{-n/2}.$$

Substitute a constant z for Δx_i such that for every x_i

$$z \leq \Delta x_i < c^{-1} \cdot 10^{-n/2}. \quad (2)$$

Then equation (??) is written

$$P_{x_i} = e^{-civ} \cdot cz \quad (3)$$

and must satisfy the condition

$$\sum_{i=0}^{\infty} cze^{-civ} = 1,$$

or

$$z = \frac{1}{c} \sum_{i=0}^{\infty} e^{-civ}.$$

But since $cv > 0$, $e^{-cv} < 1$, so that

$$\sum_{i=0}^{\infty} (e^{-cv})^i = \frac{1}{1 - e^{-cv}}$$

and finally

$$z = \frac{1 - e^{-cv}}{c}.$$

Now from ??

$$\frac{1 - e^{-cv}}{c} < \frac{10^{-n/2}}{c}.$$

Therefore

$$0 < (1 - e^{-cv}) < 10^{-n/2},$$

then

$$(1 - 10^{-n/2}) < e^{-cv} < 1$$

Thus, for $cv > 0$ we have $e^{-cv} < 1$, and for $cv < -\log(1 - 10^{-n/2})$ we have

$$-\log(1 - 10^{-n/2}) = 10^{-n/2} + \frac{10^{-(n/2)^2}}{2} + \frac{10^{-(n/2)^3}}{3} + \frac{10^{-(n/2)^4}}{4} + \dots$$

and

$$10^{-n/2} < -\log(1 - 10^{-n/2}).$$

In order that $e^{-cv} > 1 - 10^{-n/2}$ it is therefore sufficient that

$$cv \leq 10^{-n/2}. \quad (4)$$

Then we may take

$$\Delta x_i = z = \frac{1 - e^{-cv}}{c} \quad (5)$$

and substitute this value in formula (??), from which we can now set up probability tables. Here is an example :

Let $d = 10$ points as mean value to be spread on a straight line segment of length $\ell = 100$ cm. We have to define x_i and P_{x_i} as a function of i , given that $(c\Delta x_i)^2 = 10^{-4}$ is considered to be negligible.

From (??), $cv = 10^{-4/2} = 0.01$ points in v . Now $c = d/\ell = 10/100\text{points/cm}$, therefore $c = 0.1$ points/cm, $v = 0.01/0.1 = 0.1$ cm, and $x_i = 0.1 \cdot icm = imm$. From (??),

$$\Delta x_i = \frac{1 - e^{-0.01}}{0.1} = (1 - 0.9905)10 = 0.0995 = 0.1\text{cm}.$$

From (1),

$$P_{x_i} = e^{-0.01i} \cdot 0.1 \cdot 0.1 = 0.01 \cdot (0.99005)^i.$$

For calculation by machine see Chapter V.

Second Law

$$f(j) dj = \frac{2}{a} \left(1 - \frac{j}{a}\right) dj.$$

Each variable (pitch, intensity, density, etc.) forms an interval (distance) with its predecessor. Each interval is identified with a segment x taken on the axis of the variable. Let there be two points A and B on this axis corresponding to the lower and upper limits of the variable. It is then a matter of drawing at random a segment within AB whose length is included between j and $j + dj$ for $0 \leq j \leq AB$. Then the probability of this event is:

$$P_j = f(j) dj = \frac{2}{a} \left(1 - \frac{j}{a}\right) dj \quad (1)$$

for $a = AB$.

Approximate Definition of this Probability for Calculation by Hand

By taking dj as a constant and j as discontinuous we set $dj = c$, $j = iv$ with $v = a/m$ for $i = 0, 1, 2, 3, \dots, m$. Equation (??) becomes

$$P_j = \frac{2}{a} \left(1 - \frac{iv}{a}\right) c. \quad (2)$$

But

$$\sum_{i=0}^m P_j = \frac{2c}{a}(m+1) - \frac{2cv}{a^2} \sum_{i=0}^m i = \frac{2c(m+1)}{a} - \frac{2cvm(m+1)}{2a^2} = 1,$$

whence

$$dj = c = \frac{a}{m+1}.$$

On the other hand P_j must be taken as a function of the decimal approximation required:

$$P_j = \frac{2}{m+1} \left(1 - \frac{i}{m}\right) \quad (n = 0, 1, 2, 3, \dots).$$

P_j is at a maximum when $i = 0$, whence $m \geq 2 \cdot 10^n - 1$; so for $m = 2 \cdot 10^n - 1$ we have $v = a/(2 \cdot 10^n - 1)$ and $dj = a/(2 \cdot 10^n)$, and (??) becomes

$$P_j = P_i = \frac{1}{10^n} \left(1 - \frac{i}{2 \cdot 10^n - 1}\right).$$

Definition of the Same Probability for Computer Calculation

We know that the computer can only draw numbers y_0 at random (of equal probability) $0 \leq y_0 \leq 1$. Using the probability law of density $P_j = f(j) \, dj$, we have for some interval x_0

$$\text{prob. } (0 \leq j \leq x_0) = \int_0^{x_0} f(j) \, dj = \frac{2x_0}{a} - \frac{x_0^2}{a^2} = F(x_0),$$

where $F(x_0)$ is the distribution function of j . But $F(x_0) = \text{prob. } (0 \leq y \leq y_0) = y_0$. Therefore

$$\frac{2x_0}{a} - \frac{x_0^2}{a^2} = y_0 \quad \text{and} \quad x_0 = a \left(1 \pm \sqrt{1 - y_0} \right),$$

and by rejecting the positive root, since x_0 must remain smaller than a , we obtain

$$x_0 = a \left(1 - \sqrt{1 - y_0} \right)$$

for all $0 \leq x_0 \leq a$.